applications of cylinders for providing a variety of fundamental mechanical motions.

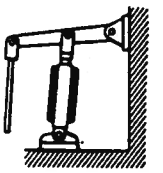
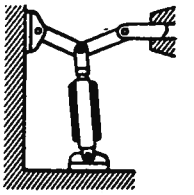
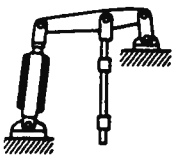
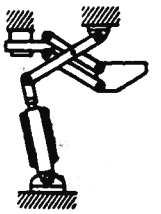
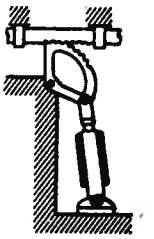
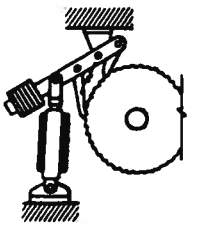
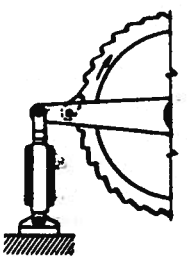
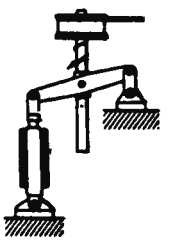
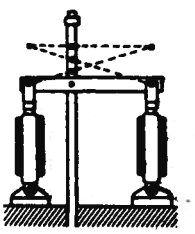
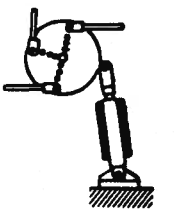
 1ST CLASS LEVER	 2ND CLASS LEVER	 3RD CLASS LEVER	 TOGGLE	 STRAIGHT LINE THRUST INCREASED
 STRAIGHT LINE MOTION MULTIPLIED 2:1	 STRAIGHT LINE MOTION IN TWO DIRECTIONS	 STRAIGHT PUSH	 HORIZONTAL PARALLEL MOTION	 PRACTICALLY CONTINUOUS ROTARY MOTION
 ENGINE BEARING	 FAST ROTARY MOTION USING STEEP SCREW NUT	 4 POSITIVE POSITIONS WITH TWO CYLINDERS	 TRAMMEL PLATE	 MOTION TRANSFERRED TO A DISTANT POINT

illustration b-10

Toggle Mechanism

For operations such as coining and marking requiring exact depth control, and requiring extremely high force for a very short distance, the toggle lever system can be useful.

CLEVIS MOUNT CYLINDER

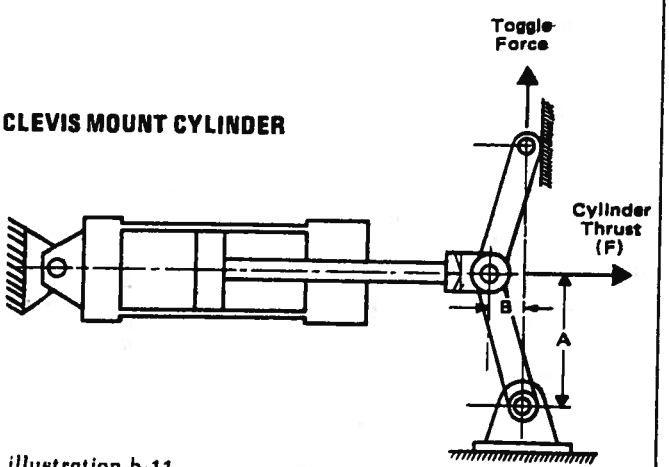


illustration b-11

In **illustration b-11,** cylinder thrust is horizontal and toggle force is taken off vertically. Bearings at each end of the toggle lever must be closely fitted and heavy enough to carry the full toggle thrust.

A calculation of toggle force can be made with the following formula, with T and F in the same units, and A and B in the same units. Note that dimension A is not the lever length, but for high leverage toggle calculations it can be used for lever length, with only small error, since the lever is nearly vertical.

$$T\ (\text{Toggle Force}) = \frac{F\ (\text{Cylinder Thrust}) \times A}{2B}$$

Example: *Find the toggle force from a cylinder thrust of 8300 lbs., if the toggle lever is 14 inches long and is 1/2 inch from vertical (Distance B).*

mechanical motion continued

Solution: *T = 8300 × 14 ÷ 2 × 1/2 = 116,200 lbs. This is a multiplication of 14 times the direct cylinder thrust. The remaining travel distance of the toggle arm at any point in the cylinder stroke is twice the difference between distance A and the true length, pin-to-pin, of the toggle arm. Distance A can be found by geometry or from a scale layout.*

Thrust Exerted by a Cylinder Working at an Angle

Cylinder thrust, F, is horizontal in this figure. Only that vector force, T, which is at right angles to the lever axis is effective for turning the lever. The value of T varies with the acute angle "A" between cylinder and lever axes.

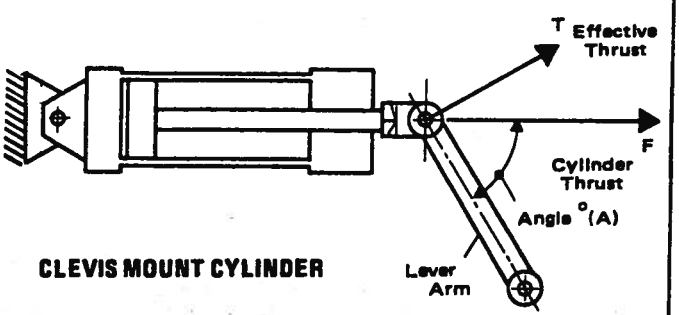


illustration b-12

To calculate T, multiply cylinder thrust times the power factor taken from **table b-11** below.

Example: *A 3.25-inch bore cylinder working 1000 p.s.i. gauge pressure will develop a 8300 lb. thrust (8.294 in. area × 1000). Effective thrust when working at a 65° angle is: 8300 × 0.906 (from table 10) = 7519.8.*

power factor table

ANGLE A DEGREES	PWR. FACTOR (SIN A)	ANGLE A DEGREES	PWR. FACTOR (SIN A)
5	0.087	50	0.766
10	0.174	55	0.819
15	0.259	60	0.867
20	0.342	65	0.906
25	0.423	70	0.940
30	0.500	75	0.966
35	0.573	80	0.985
40	0.643	85	0.996
45	0.707	90	1.000

table b-11

Determining Cylinder Stroke for Operating a Hinged Lever by the Chord Factor Method

If the cylinder is rotating the lever to an equal angle each side of the perpendicular as in **illustration b-13,** the length of stroke can very easily be determined by multiplying lever length (pin-to-pin) times the chord factor from **table b-12.** If the movement is not equal on each side of the perpendicular, the stroke may be determined by using another example shown in this section.

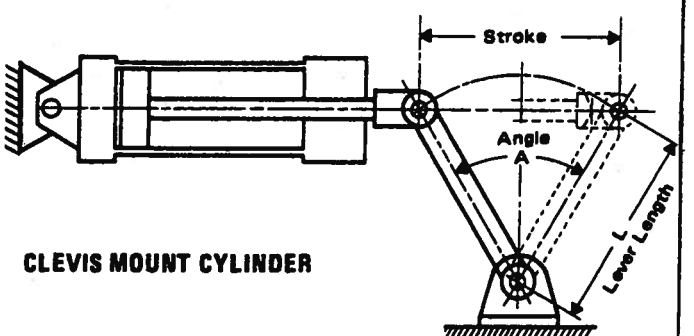


illustration b-13

Example: *The cylinder stroke needed to swing a 15-inch lever through a 120 degree arc, when mounted as in* **illustration b-12,** *is found by taking the factor 1.732 (from* **table b-12***) times 15 (lever length) = 25.98" (stroke length).* Many times a stock cylinder with standardized stroke length can be used by lengthening or shortening the lever arm for the desired travel.

ANGLE A DEGREES	CHORD FACTOR	ANGLE A DEGREES	CHORD FACTOR
5	0.087	85	1.351
10	0.174	90	1.414
15	0.261	95	1.475
20	0.347	100	1.532
25	0.433	105	1.587
30	0.518	110	1.638
35	0.601	115	1.687
40	0.684	120	1.732
45	0.765	125	1.774
50	0.845	130	1.813
55	0.923	135	1.848
60	1.000	140	1.879
65	1.075	145	1.907
70	1.147	150	1.932
75	1.217	155	1.953
80	1.286	160	1.970

table b-12

mechanical motion continued

Cranes and Beams

Because crane working angles constantly change, constructing a rough model on paper is often necessary to show the point at which the greatest cylinder thrust is needed. An exact calculation can then be made.

Only the portion of cylinder thrust at right angles to the beam axis is effective for turning the beam. This can be calculated by the method shown below.

Example: *Find the maximum load that can be lifted by the crane when the angles are as shown, and the cylinder thrust, F, is 19635 lbs. Translate the 19635 lbs. cylinder thrust into F_1, 9817.5 lbs. at right angles to the beam, using the power factor of 0.500 lbs. for a 30° angle from* **table b-12, page b-19.** *Next, translate this to F_2, 3272.5 lbs. thrust at the weighted end of the beam. This is done with simple proportion by the length of each arm from the base pivot point. F_2 is 1/3 F_1, since the lever is 3 times as long. Then, using the power factor* **table b-12,** *find the maximum hanging load that can be lifted at a 45° angle between beam and load weight.*

Calculations for Heavy Beam

With heavy beams, it is necessary to take into account the weight of the beam itself. If the beam is distributed uniform in weight across its length, the calculation is relatively easy. In **illustration b-15** the beam has a uniform weight of 90 pounds per foot. It is partially counter balanced by a load weight of 400 pounds on the left side of the fulcrum, and must be raised by the cylinder force applied at a point 9 feet from the right side of the fulcrum.

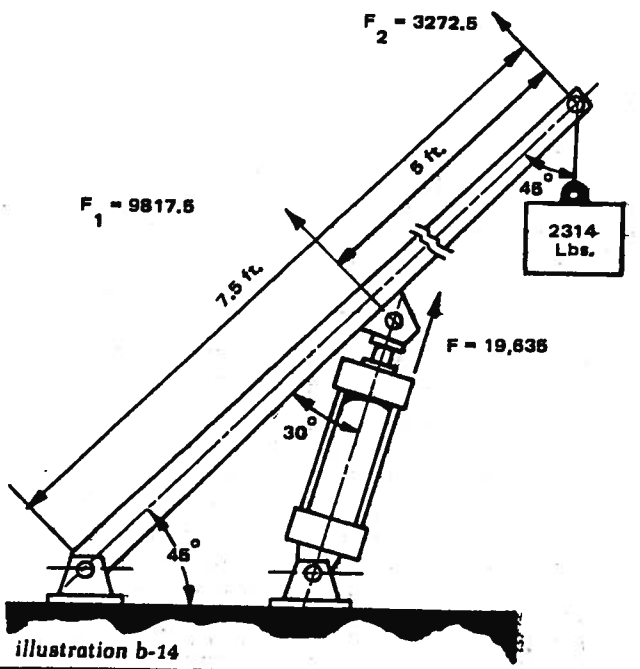


illustration b-14

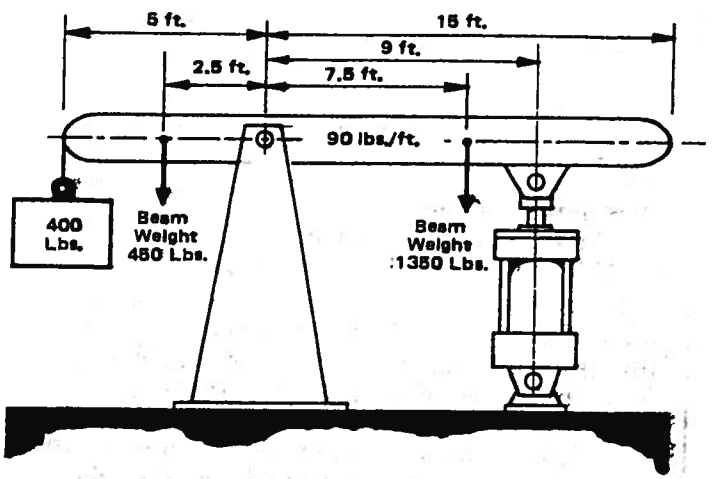


illustration b-15

A method of solution is to use the principle of moments. A moment is a torque force consisting of (so many) pounds applied at a lever distance of (so many) feet or inches. The solution here is to find how much cylinder thrust is needed to just balance the beam. Then by increasing the hydraulic cylinder thrust about 10 to 15% to take care of friction losses, the cylinder would be able to raise the beam.

Using the principle of moments, it is necessary to calculate all of the moment forces which are trying to turn the beam clockwise, then calculate all the moment forces trying to turn the beam counter-clockwise, then subtract the two. In this case they must be equal to balance the beam.

Clockwise moment due to the 15 feet of beam on the right side of the fulcrum: This can be considered as a concentrated weight acting at its center of gravity 7-1/2 feet from the fulcrum. Moment = 90 (lbs. per foot) × 15 feet × 7-1/2 feet = 10125 foot pounds.

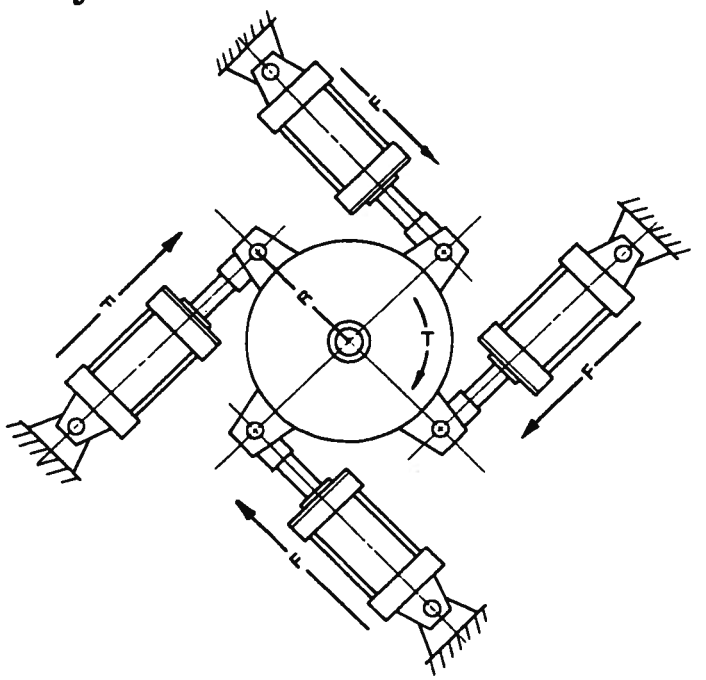
Counter-clockwise moment due to the 5 feet of beam on the left side of the fulcrum: 90 (lbs. per foot) × 5 feet × 2-1/2 feet (CG distance) = 1125 foot pounds.

Counter-clockwise moment due to hanging

continued on page b-24

mechanical motion continued

rotary motion



Formula:

$$\text{Torque} = 4F \times R$$

F = Force
R = Radius
T = Torque

Problem:

Torque required = 2000 lb. ft.
R = .5 ft.
T = 4F x R

$$F = \frac{T}{4R}$$

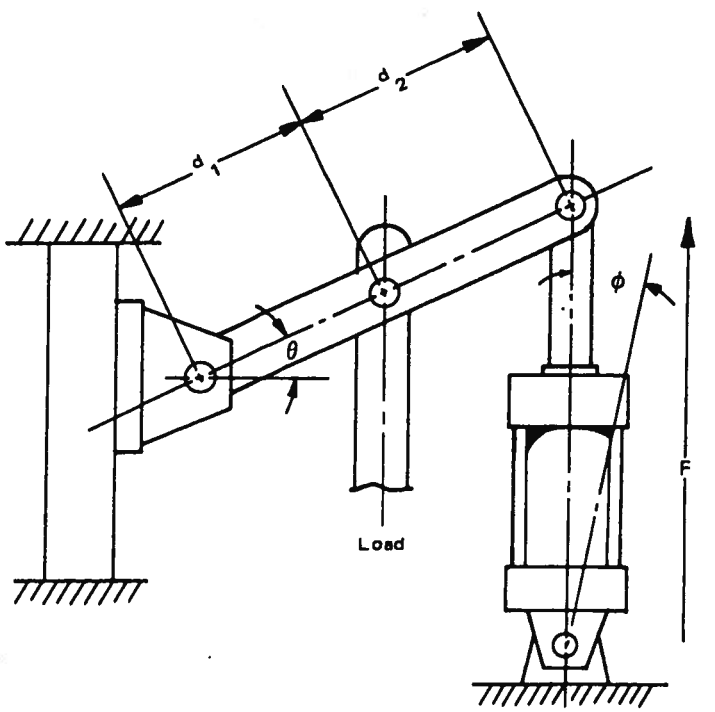
$$F = \frac{2000 \text{ lb. ft.}}{4 \times .5}$$

F = 1000 lb. Force

Use 1-1/2 inch bore cylinders operating at 600 psi.

illustration b-16

second class lever



Formula:

$$F\cos\phi \times (d_1 + d_2)\cos\theta = \text{Load} \times d_1\cos\theta$$

$$F\cos\phi\,(d_1 + d_2) = \text{Load} \times d_1$$

$$L = \frac{F\cos\phi\,(d_1 + d_2)}{d_1}$$

For ϕ less than + or − 15°, $\cos \approx 1$

$$\therefore \text{Load} = \frac{F(d_1 + d_2)}{d_1}$$

ϕ = angle between the vertical and the line of action of the cylinder.

Problem:

L = 4000 lb.
d_1 = 10 in.
d_2 = 20 in.

$$L = \frac{F(d_1 + d_2)}{d_1}$$

$$F = \frac{Ld_1}{d_1 + d_2}$$

$$F = \frac{4000 \times 10}{30}$$

F = 1333 lb. Force

Select a pressure of 500 psi.

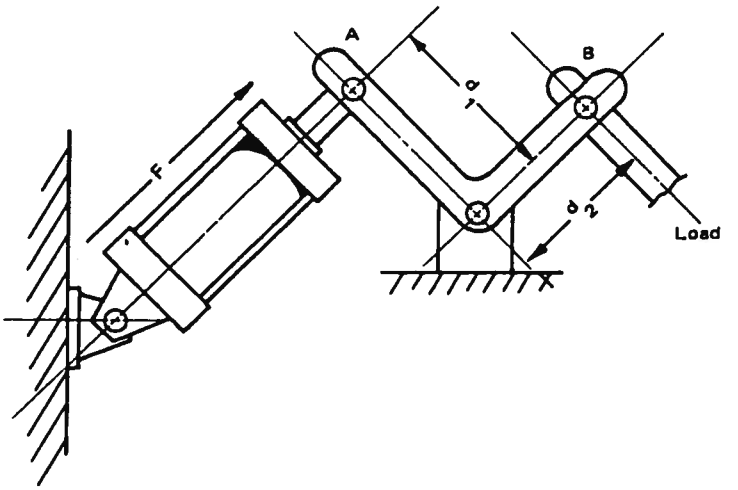
$$F = P \times A$$
$$1333 = 500 \times A$$
$$A = \frac{1333}{500} = 2.7 \text{ in.}^2$$

Use standard 2 in. bore cylinder.

illustration b-17

mechanical motion continued

lever



Formula:

$$Fd_1 = Ld_2$$

$$L = \frac{Fd_1}{d_2}$$

As long as the applied force is less than 15° from the normal to A.

Problem:

d_1 = 8 in.
d_2 = 12 in.
L = 10,000 lb.

$$L = \frac{Fd_1}{d_2}$$

F = Force

$$F = \frac{Ld_2}{d_1}$$

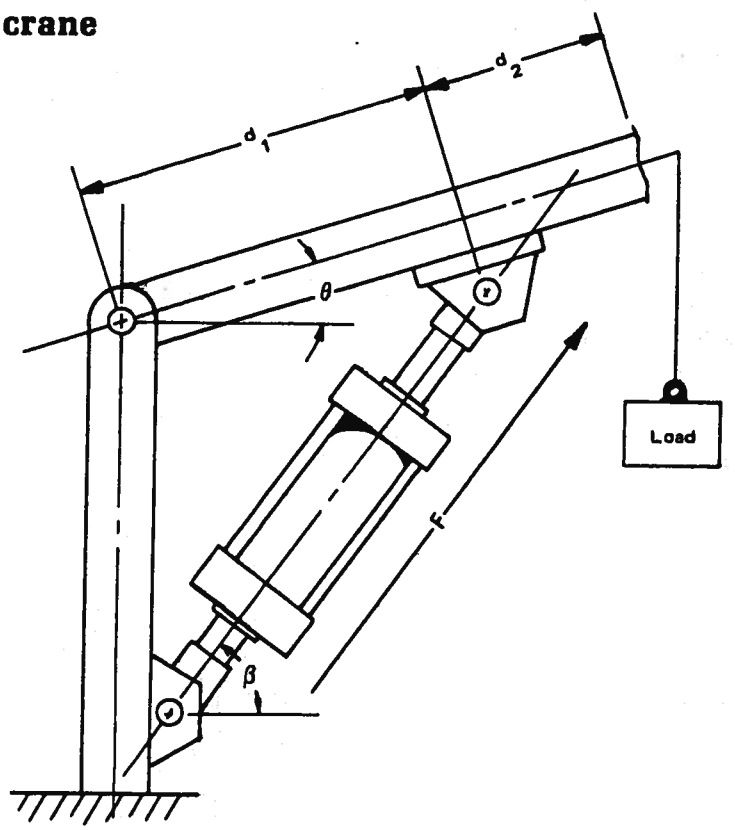
$$F = \frac{10{,}000 \times 12}{8}$$

F = 15,000 lb. Force

Select 4 in. bore cylinder with cross sectional area of 12 in.^{2} operating at less than 1500 psi.

illustration b-18

crane



Formula:

$$F\sin\beta \times d_1\cos\theta = L(d_1 + d_2)\cos\theta$$

$$F\sin\beta \times d_1 = L(d_1 + d_2)$$

$$L = \frac{F\sin\beta \times d_1}{d_1 + d_2}$$

Where β in minimum angle between center line of cylinder and the horizontal.

Problem:

d_1 = 15 in.
d_2 = 60 in.
L = 1000 lb.
β = 45°

$$L = \frac{F\sin\beta \times d_1}{d_1 + d_2}$$

F = Force

$$F = \frac{L(d_1 + d_2)}{\sin\beta \times d_1}$$

$$F = \frac{1000\ (75)}{.707 \times 15}$$

F = 7071 lb. Force

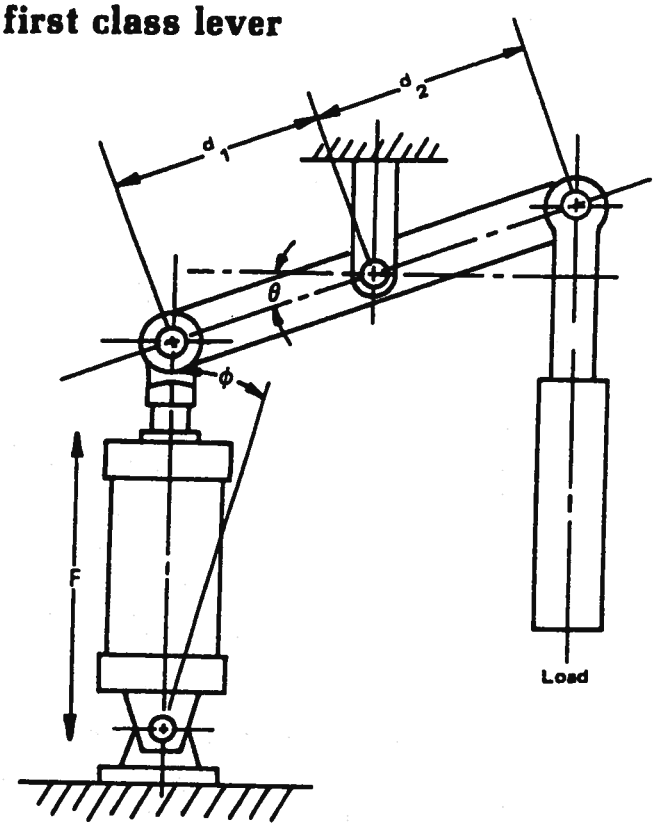
Cylinder operating at 1000 psi. would require an area of 7 in.^{2}

Use standard 3 1/4 in. bore cylinder.

illustration b-19

mechanical motion continued

first class lever



Formula:

$$F \cos\phi \times d_1 \cos\theta = \text{Load} \times d_2 \cos\theta$$

$$F \cos\phi \times d_1 = L \times d_2$$

$$L = \frac{Fd_2}{d_2}$$

Where 0 is between +15→ −15 of the vertical, $\cos\phi$ is approximately = 1

$$\therefore L = \frac{Fd_1}{d_2}$$

Problem:

2 1/2 in. bore cylinder
d_1 = 5 in.
d_2 = 10
θ = 30°
L = 1800 lb.

$$L = \frac{Fd_1}{d_2}$$

$$Ld_2 = Fd_1$$

$$F = \frac{Ld_2}{d_1}$$

$$F = \frac{1800 \times 10}{5}$$

$$F = 3600 \text{ Force}$$

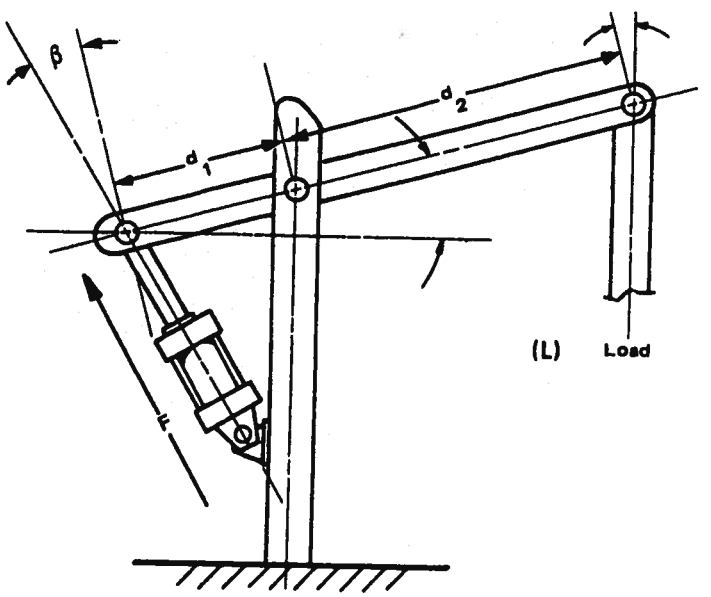
$$F = P \times A$$

$$3600 = P \times 4.9$$

$$P = \frac{3600}{4.9} = 750 \text{ psi.}$$

illustration b-20

first class lever variation



Formula:

$$F \cos\beta \times d_1 \cos\theta = L \times d_2 \cos\phi$$

$$F \cos\beta \times d_1 = L \times d_2$$

$$L = \frac{F \cos\beta \times d_1}{d_2}$$

Problem:

L = 800 lb. Force
β = 20°
d_1 = 10 in.
d_2 = 30 in.

$$\gamma = \frac{F \cos\beta \times d_1}{d_2}$$

$$F = \frac{L \times d_2}{\cos\beta \times d_1}$$

$$F = \frac{800 \times 30}{.939 \times 10}$$

$$F = 2554 \text{ lb. Force}$$

Use 2 1/2 in. bore cylinder operating at approximately 500 psi.

illustration b-21

mechanical motion continued

toggle variation

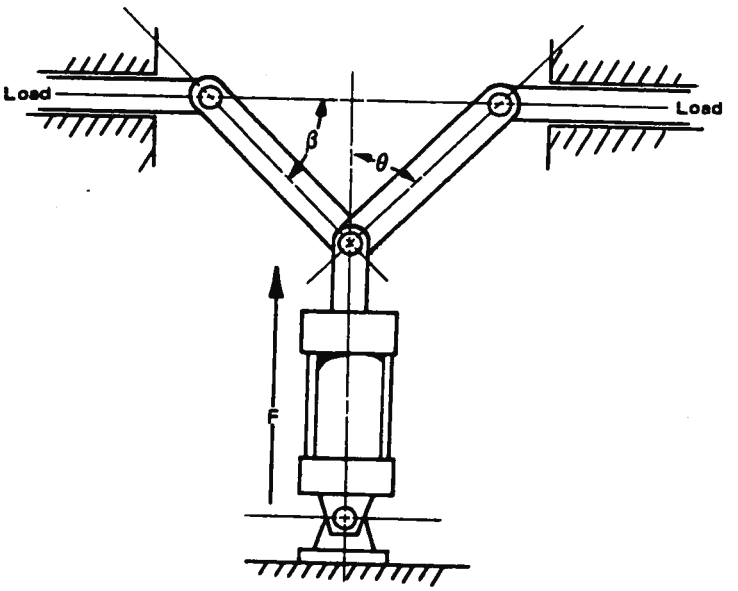


illustration b-22

Formula:

$$\text{Load} = \frac{F}{2\cos\theta} \sin\beta$$

$$\text{Load}_1 = \text{Load}_2$$

θ between 30° – 60°
β between 30° – 60°

Problem:

Load is 500 pounds on both sides, therefore, total load is 1000 pounds.

$\theta = 45°$
$\beta = 45°$

$$L = \frac{F\sin\beta}{2\cos\theta}$$

$$F = \frac{L2\cos\theta}{\sin\beta}$$

$$F = \frac{2 \times 1000 \times .7}{.7} = 2000 \text{ lb. force}$$

Use 1 in. bore cylinder with a cross sectional area of about 1 in.^{2} operating at less than 1000 psi.

weight of 400 pounds: 400 lbs. × 5 feet = 2000 foot pounds.

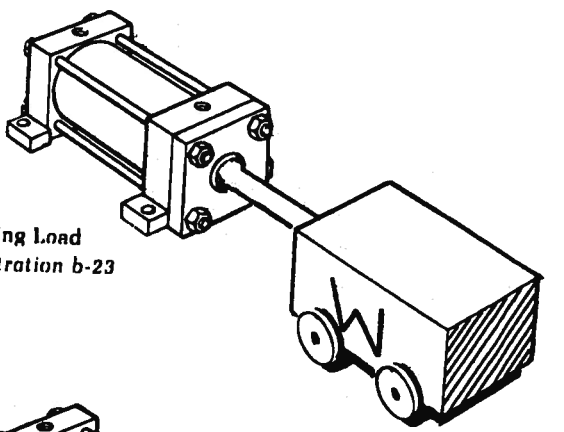
Subtracting counter-clockwise from clockwise moments: 10125 - 1125 - 2000 = 7000 foot pounds that must be supplied by the cylinder for balance conditions. To find cylinder thrust: 7000 foot pounds ÷ 9 feet (distance from fulcrum) = 777.77 pounds.

Note: *When working with moments, that only the portion of the total force which is at right angles to the beam is effective as a moment force. If the beam is at an angle to the cylinder or to the horizontal, then the effective portion of the concentrated or distributed weight, and the cylinder thrust, can be calculated with power factors by the method shown in this section.*

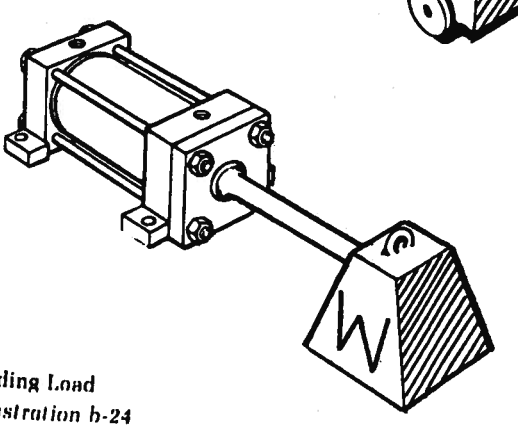
Cylinders Moving Horizontal Loads

Rolling Loads — A cylinder thrust of 1/10 the load weight will move loads which operate on low-friction needle, roller, or ball bearings.

An air cylinder with meter-out flow controls can be used on some applications, even at slow feed rates. Attention should be given to deceleration at the end of cylinder stroke to



Rolling Load
illustration b-23



Sliding Load
illustration b-24

prevent momentum of the load from damaging the cylinder and/or the machine.

Sliding Loads — Either air or hydraulic cylinders can be used for moving high friction sliding loads. (**illustration b-24, page b-24**).

On applications that require rapid indexing from one positive stop to another, an air cylinder will give more rapid action than hydraulics if the load is within its capacity. Air cylinders should **not** be used for slow speed or controlled feeding of a sliding load with a large area of surface friction, as a chattering motion will result. Hydraulic cylinders with a meter-out flow control should be used in these applications. In some instances an air/oil system will also give acceptable performance.

The force needed to push a sliding load varies with surface material, lubrication, unit loading and other factors. For lightly lubricated machined slides, the cylinder thrust should be equal to 1/2 to 3/4 of the load weight to get the load started. A thrust of 1/5 to 1/6 load weight will keep it moving.

To operate an air cylinder at high speeds, the cylinder should be sized to develope twice the thrust needed to balance the load.

Cylinders for Lifting

Differential Lift — Where overhead clearance is not sufficient for a direct lift, a differential lift is sometimes an ideal solution. A shorter length, larger diameter cylinder is usually best in this application.

The arrangement illustrated will give a 2:1 mechanical reduction. Size the cylinder with twice the piston area and half the stroke needed for a straight lift of the same load.

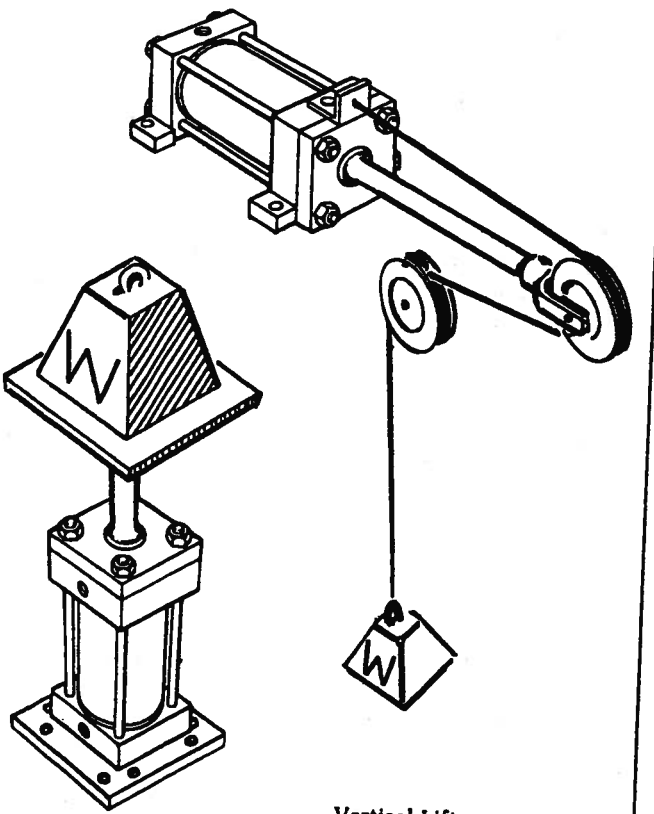
Possible side thrust from the cylinder is prevented by running the pulley attached to the cylinder rod in horizontal guides.

In addition to needing less head room, this arrangement allows the cylinder to work on full piston area, and the rod packings are not subjected to high pressure.

Vertical Lifting — Air cylinders used to lift a load must always be sized to exert a force greater than the weight of the load. An air cylinder which exerts a 500 lb. force can support a 500 lb. load, but cannot move it. For normal applications, an air cylinder should develop 25% more thrust than needed to support the load. Twice the force needed to support the load is required for fast operation.

A hydraulic or air/oil system must be used if

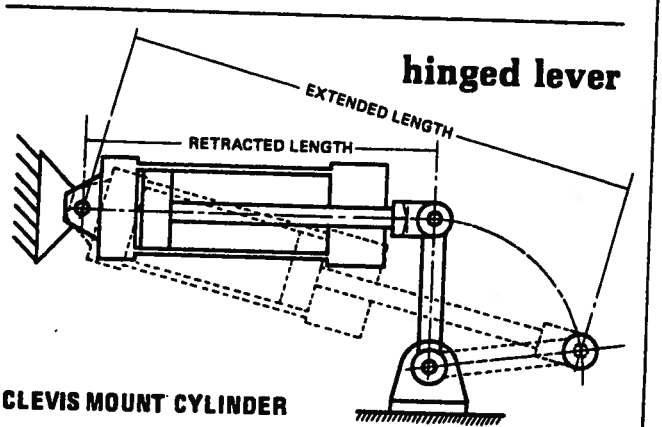
Differential Lift



Vertical Lift

illustration b-25

hinged lever



CLEVIS MOUNT CYLINDER

illustration b-26

the cylinder is to be stopped at an intermediate point for loading or unloading.

Determining Cylinder Stroke for Operating a Hinged Lever Pushing at an Angle by the Scale Layout Method

In all cases a sketch should be made, showing the length and angular travel of the lever, and